



Solar Carports, Enhanced Charging Technology with SiC (Credit: Diotec Semiconductor)

SiC devices, optimises switching behaviour further by minimising the parasitic inductance usually found in traditional setups.

The adoption of silicon carbide technology in electric vehicles is a game-changer for the automotive industry. It not only improves the performance and efficiency of the vehicles but also contributes to the broader goal of reducing carbon emissions. As EVs continue to grow in popularity, innovations like SiC are critical in making sustainable transportation a practical reality for more people around the world.

The integration of silicon carbide into electric vehicle technology represents a significant leap forward in our journey towards an electrified, efficient, and environment friendly transportation system. As this technology continues to evolve and scale, it promises to unlock new possibilities for the automotive industry, driving us towards a more sustainable future. Silicon carbide is not just an alternative; it is becoming the standard for the next generation of electric vehicles.

Amid the surging demand for electric vehicles, the need for efficient and robust onboard chargers

has become a top priority. Silicon carbide technology is becoming a cornerstone in this development, thanks to its superior properties compared to traditional silicon in power applications. This shift promises to significantly boost the performance and efficiency of electric vehicle chargers, a key component in the EV infrastructure.

Automotive qualifications

A key highlight in the utilisation of SiC technology is its compliance with the rigorous AEC-Q101 standard, which is critical for automotive applications. This certification ensures that SiC devices can perform under the demanding conditions typical in automotive environments, particularly in power factor correction (PFC) circuits used in EV chargers.

The move towards silicon carbide is not just about enhancing current technologies but is also a step towards futureproofing the automotive industry. As EVs continue to evolve, the requirements for more efficient, reliable, and compact power solutions grow. Silicon carbide stands out as a material that can meet these needs today and tomorrow.

By adopting SiC in their products, companies are not only addressing the immediate needs of the automotive sector but are also setting new standards for the performance of electronic components in EVs. This collaboration is poised to drive significant advancements in the industry, reducing carbon footprints and enhancing the sustainability of vehicle technologies. This development not only enhances the efficiency and performance of

electric vehicles but also aligns with global efforts to promote sustainable transportation. As silicon carbide technology continues to mature, its role in the automotive industry is set to become more pivotal, promising a cleaner, more efficient future for all.

SiC: Reliability, efficiency, and manufacturing excellence

As the electric vehicle (EV) industry evolves, manufacturers continue to seek out innovative solutions that enhance performance, reliability, and lifespan. Silicon carbide, a robust semiconductor material, is emerging as a key player in this transformation, offering significant advantages over traditional silicon in high-power applications.

Silicon carbide stands out for its high thermal conductivity and ability to operate at lower temperatures, which significantly enhances its reliability. This makes SiC an ideal choice for demanding environments, particularly in the automotive industry, where high quality and reliability are paramount. Vehicles equipped with SiC-based devices can enjoy longer lifespans and reduced maintenance costs, making SiC a preferred material for